# Digikala Online Shop Simulator

Mahla Entezari — Mahla.Entezarii@gmail.com

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## 1 Introduction

This project is an online shop simulator inspired by platforms such as Digikala and Amazon. It simulates a basic e-commerce experience using Java, JavaFX for GUI, and MySQL for the backend database. It supports functionalities such as user authentication, product browsing, cart and order handling, and comment management.

## 2 Technologies Used

- Java (JavaFX)
- MySQL
- FXML (JavaFX Scene Builder)
- MVC (Model-View-Controller) Architecture

## 3 System Overview

The system comprises several Java classes that handle different responsibilities:

- **Main.java:** Launches the JavaFX application and initializes the database.
- **ConnectDB.java:** Handles database connection and SQL query execution.
- **DigikalaService.java:** Acts as a backend service for user management, product handling, and login logic.
- **User.java, Admin, Seller:** Represent the user model and its variants.
- **Login.java, Createaccount.java, HelloController.java:** Handle frontend interactions for login, account creation, and navigation.
- **UserPage.java:** Displays categorized product listings and dynamically updates the UI.

# 4 Database Integration

A MySQL database is used to store products and user-related data. The application executes queries to fetch products by category and updates the UI accordingly.

# 5 Key Functionalities

- **Login System:** Allows login for three roles – user, admin, and seller.
- **Product Display:** Products are fetched from the database and shown under categories like Mobile, Laptop, TV, etc.
- **Shopping Cart:** Users can add products to their cart and view their orders.
- **Account Management:** Users can register new accounts.
- **Comment System:** Comments are stored in a dedicated table and shown for each product.

# 6 Challenges Faced

- GUI binding with the backend logic.
- Displaying products dynamically in JavaFX components.
- Designing a flexible database schema for product comments.

# 7 Bonus Implementations

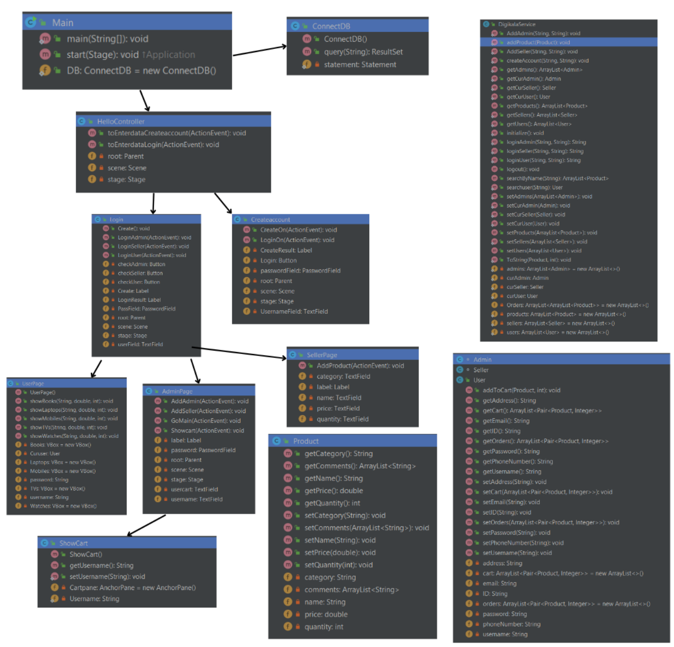
1. Graphical User Interface (JavaFX)
2. Commenting system for products
3. Search functionality
4. Persistent data storage via MySQL
5. Order and sales history
6. UUIDs for user identification

# 8 Sample Code

Here is an example of database connectivity:

```
public class ConnectDB {
    static private Statement statement;
    public ConnectDB() {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection connection = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/jdbc", "root", "12345678");
            statement = connection.createStatement();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

# 9 UML Diagram




# 10 Conclusion

This project successfully simulates a simple online shop system with all necessary modules including user login, product handling, and persistent storage. The integration of JavaFX and MySQL enables a smooth and interactive user experience.

# 11 Resources

- JavaFX Course
- MySQL Course